

# DES & Feistel Network Architecture

## 1. Feistel Cipher Round Diagram

In a Feistel cipher, the plaintext is split into two halves. The right half is passed through a round function and then XORed with the left half.

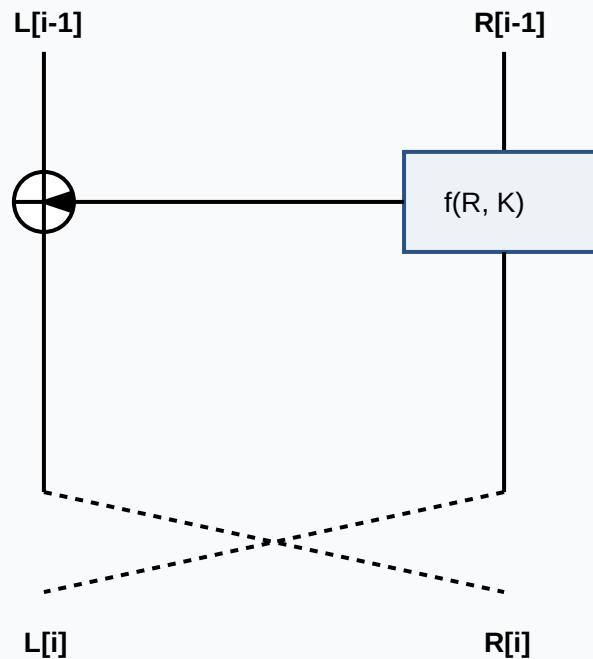


Figure 1: Logic of a single Feistel encryption round

$$\begin{aligned} L_i &= R_{i-1} \\ R_i &= L_{i-1} \oplus f(R_{i-1}, K_i) \end{aligned}$$

## 2. S-Box Internal Mechanism

The S-box is the core non-linear component of DES. It maps a 6-bit input to a 4-bit output to provide confusion.

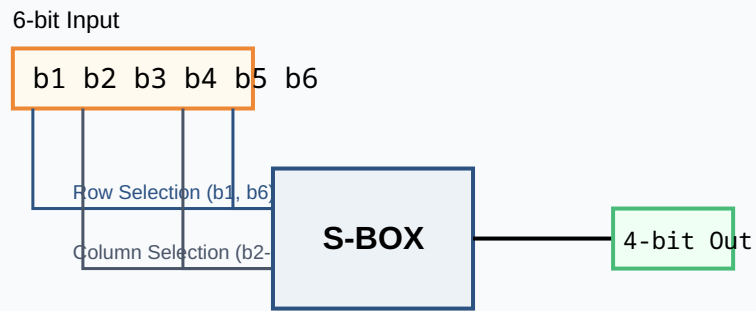


Figure 2: DES S-Box Bit Mapping (Selection Logic)

## Key Properties:

- **Non-linearity:** Ensures the cipher cannot be broken by linear cryptanalysis.
- **Confusion:** Makes the relationship between the key and ciphertext complex.
- **Compression:** It reduces the 48-bit intermediate key-mixed data back down to 32 bits for the next permutation.